

Mark Sui

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Education

University of California, San Diego

Master of Science in Electrical and Computer Engineering

- Coursework: VLSI, Computer Architecture, FPGA/ASIC Design, Machine Learning

Sep 2025 – Jun 2026

GPA: 3.6/4.0

University of California, San Diego

Bachelor of Science in Computer Engineering

Bachelor of Science in Mathematics (Double Major)

- Coursework: Digital Circuits and Design, Data Structures, Embedded Systems, Deep Learning

Sep 2021 – Jun 2025

GPA: 3.6/4.0

Experience

Research Fellow · Handshake AI (Remote)

Jun 2025 – Dec 2025

- Built Python evaluation pipelines and regression frameworks for agentic systems to run structured test suites, capture step-level execution traces, and generate reproducible technical readouts.
- Analyzed **10,000+** execution traces to classify tool misuse, planning errors, and divergence into repeatable failure taxonomies.

Embedded Systems Engineer Intern · Shandong Haike Information Technology (Remote)

Jun 2024 – Dec 2024

- Integrated mmWave radar modules into a camera-free IoT health monitoring system; instrumented real-time hardware data logging and module calibration.
- Validated sensing behavior across distance, orientation, and reflective surfaces, isolating weak-signal and stability failure modes.

Software Developer Intern · Liqun Group (Qingdao, China)

Jun 2023 – Sep 2023

- Implemented secure backend logic and APIs; optimized data matching similarity weights to improve response latency by **15%**.

Selected Technical Projects

Dual-Core Systolic Attention Accelerator

RTL to PnR

- Microarchitecture & RTL Design: Microarchitected a dual-core 8×8 INT8 vector MAC matrix engine in synthesizable Verilog; implemented control FSMs, SRAM-backed ping-pong memories, output FIFOs, and asynchronous FIFO-based CDC paths.
- Verification & CDC Analysis: Developed Python golden models and directed testbenches to validate multi-domain clock gating, reset synchronization, and boundary conditions under high-throughput dense/sparse matrix workloads.
- Physical Implementation: Executed complete synthesis and PnR flow with SRAM macros; drove timing closure to recover setup WNS from **-4.22 ns to -0.87 ns** via adder-tree logic balancing, pipeline retiming, and strict CDC constraints.

Convolutional Viterbi Decoder

FPGA/RTL

- Digital Core Implementation: Designed a rate-1/2 convolutional transceiver system in SystemVerilog, incorporating an 8-state Viterbi decoder with optimized Branch Metric Computation (BMC), 8 Add-Compare-Select (ACS) units, and pipelined memory bank rotation for traceback control.
- Validation & Prototyping: Built automated bit-error injection scoreboards to verify deterministic, random, and burst error recovery; synthesized and hardware-validated the design on Intel FPGA via Quartus, optimizing resource footprint to **361 ALUTs and 239 registers**.

MAGE RTL Debugging and VerilogEval Flow

AI/EDA, Verilog

- Multi-Agent LLM Framework: Frameworked an automated RTL generation and debugging flow for VerilogEval-V2, integrating simulator-grounded feedback loops and an error-specific routing mechanism (syntax/interface/logic failures) with task-local state checkpointing.
- Automated Triage & Evaluation: Benchmarked **148** paired Verilog designs to profile LLM regressive-edit behaviors; engineered automated triage tools and JSON-backed run-level reports to classify hardware failure modes, boosting logic repair efficiency.

Croc SoC Physical Design and PPA Optimization

OpenROAD/PPA

- Automated DSE Flow: Developed Shell/Tcl scripts to enable parameterized floorplanning and macro topologies; swept core aspect ratios, cell utilization, and routing margins to achieve multi-dimensional design closure.
- PPA Optimization: Reduced clock network power by **19.1%**, total power by **2.8%**, and area by **0.60%** via downsized CTS buffers, zero-margin hold repairs, and power recovery, achieving **zero WNS/TNS/DRC violations** across selected runs.
- Hierarchical Handoff: Engineered a bottom-up hierarchical flow for core and peripheral macros; generated abstract LEF/Liberty timing models and successfully integrated hard macros into the top-level floorplan.

Skills

RTL Design & Microarchitecture: Verilog, SystemVerilog, FSM/Control Logic, Pipelined Datapaths, Fixed-Point Arithmetic, SRAM/FIFO Interfaces, Clock Domain Crossing (CDC), Reset Synchronization

Verification & Prototyping: Universal Verification Methodology (UVM) concepts, SystemVerilog Assertions (SVA), Directed/Random Testbenches, Python Golden Models, Waveform Debug, Quartus, ModelSim, Xcelium, Yosys

EDA, STA & Physical Implementation: Synthesis, Static Timing Analysis (STA), OpenROAD/OpenDB, Setup/Hold Closure, Critical-Path Analysis, WNS/TNS Recovery

Programming & Automation: Python, C/C++, CUDA, Tcl, Bash, Make, Linux, Git, MATLAB, NumPy, PyTorch